

Module 3 — Query Results

PostgreSQL analysis of the Grad Café applicant dataset

Question 1

How many entries applied for Fall 2026?

Result: 29,576

SQL query

```
SELECT COUNT(*)
FROM applicants
WHERE LOWER(term) = 'fall 2026';
```

Explanation: Counts all applicant records whose term is Fall 2026. LOWER() makes the text comparison insensitive to capitalization.

Question 2

What percentage of applicants are international?

Result: 46.34%

SQL query

```
SELECT
    100.0 * COUNT(*) FILTER (
        WHERE LOWER(us_or_international) = 'international'
    )
    / NULLIF(COUNT(*) FILTER (
        WHERE us_or_international IS NOT NULL
        AND TRIM(us_or_international) <> ''
    ), 0)
FROM applicants;
```

Explanation: Divides the number classified as International by the number with a nonblank nationality classification, then multiplies by 100. Missing or blank classifications are excluded from the denominator.

Question 3

What are the average GPA and GRE scores?

Result: Average GPA: 3.79; Average GRE Quantitative: 259.81; Average GRE Verbal: 161.53; Average GRE Analytical Writing: 8.34

SQL query

```
SELECT
    AVG(gpa),
    AVG(gre),
    AVG(gre_v),
    AVG(gre_aw)
FROM applicants;
```

Explanation: Calculates each average independently. PostgreSQL AVG() ignores NULL values, so each metric uses only records that contain that metric. The source data are self-reported and include implausible GRE values, so these averages reflect the stored dataset and should be interpreted cautiously.

Question 4

What is the average GPA of American applicants for Fall 2026?

Result: 3.79

SQL query

```
SELECT AVG(gpa)
FROM applicants
WHERE LOWER(term) = 'fall 2026'
AND LOWER(us_or_international) = 'american'
AND gpa IS NOT NULL;
```

Explanation: Filters to American applicants in Fall 2026 with a reported GPA, then calculates their average GPA.

Question 5

What percentage of Fall 2025 applications were accepted?

Result: 47.92%

SQL query

```
SELECT
    100.0 * COUNT(*) FILTER (
        WHERE LOWER(status) = 'accepted'
    )
    / NULLIF(COUNT(*), 0)
FROM applicants
WHERE LOWER(term) = 'fall 2025';
```

Explanation: Restricts the data to Fall 2025 and divides accepted applications by all Fall 2025 applications. NULLIF prevents division by zero.

Question 6

What is the average GPA of accepted applicants for Fall 2026?

Result: 3.78

SQL query

```
SELECT AVG(gpa)
FROM applicants
WHERE LOWER(term) = 'fall 2026'
AND LOWER(status) = 'accepted'
AND gpa IS NOT NULL;
```

Explanation: Filters to accepted Fall 2026 applicants with a reported GPA and calculates their average GPA.

Question 7

How many applicants applied to a Johns Hopkins master's program in Computer Science?

Result: 8

SQL query

```
SELECT COUNT(*)
FROM applicants
WHERE (
    LOWER(program) LIKE '%johns hopkins%'
    OR LOWER(program) LIKE '%jhu%'
)
AND (
    LOWER(program) LIKE '%computer science%'
    OR LOWER(program) LIKE '%cs%'
)
AND (
    LOWER(degree) LIKE '%master%'
    OR LOWER(degree) IN ('ms', 'm.s.', 'msc', 'm.sc.')
);
```

Explanation: Uses the original combined university/program text and degree field. It allows common variations of Johns Hopkins, Computer Science, and master's degree labels.

Question 8

How many Fall 2026 accepted applicants applied for a PhD in Computer Science at Georgetown, MIT, Stanford, or Carnegie Mellon?

Result: 28

SQL query

```
SELECT COUNT(*)
FROM applicants
WHERE LOWER(term) = 'fall 2026'
  AND LOWER(status) = 'accepted'
  AND LOWER(degree) IN ('phd', 'ph.d.', 'ph.d')
  AND (
    LOWER(program) LIKE '%computer science%'
    OR LOWER(program) LIKE '% cs%'
  )
  AND (
    LOWER(program) LIKE '%georgetown%'
    OR LOWER(program) LIKE '%massachusetts institute of technology%'
    OR LOWER(program) LIKE '% mit,%'
    OR LOWER(program) LIKE '%stanford%'
    OR LOWER(program) LIKE '%carnegie mellon%'
    OR LOWER(program) LIKE '%cmu%'
  );
```

Explanation: Uses the original downloaded university/program information to count accepted Fall 2026 Computer Science PhD applicants at the four specified universities, while allowing common name abbreviations.

Question 9

Repeat Question 8 using the LLM-generated program and university fields.

Result: Original-field count: 28; LLM-generated-field count: 24; Difference: -4

SQL query

```
SELECT COUNT(*)
FROM applicants
WHERE LOWER(term) = 'fall 2026'
  AND LOWER(status) = 'accepted'
  AND LOWER(degree) IN ('phd', 'ph.d.', 'ph.d')
  AND (
    LOWER(llm_generated_program) LIKE '%computer science%'
    OR LOWER(llm_generated_program) = 'cs'
  )
  AND (
    LOWER(llm_generated_university) LIKE '%georgetown%'
    OR LOWER(llm_generated_university) LIKE '%massachusetts institute of technology%'
    OR LOWER(llm_generated_university) = 'mit'
    OR LOWER(llm_generated_university) LIKE '%stanford%'
    OR LOWER(llm_generated_university) LIKE '%carnegie mellon%'
    OR LOWER(llm_generated_university) = 'cmu'
  );
```

Explanation: The LLM-generated fields return 24 records, four fewer than the original-field query. Standardization can change the wording of university or program names, which can change which records satisfy exact LIKE-based matching rules.

Question 10

How many Fall 2026 applicants were accepted?

Result: 10,973

SQL query

```
SELECT COUNT(*)
FROM applicants
WHERE LOWER(term) = 'fall 2026'
      AND LOWER(status) = 'accepted';
```

Explanation: Counts records that are both Fall 2026 and Accepted.

Question 11

What is the average GPA of international applicants for Fall 2026?

Result: 3.79

SQL query

```
SELECT AVG(gpa)
FROM applicants
WHERE LOWER(term) = 'fall 2026'
      AND LOWER(us_or_international) = 'international'
      AND gpa IS NOT NULL;
```

Explanation: Filters to international Fall 2026 applicants with a reported GPA and calculates the average.